

Experimental Setup

Does the proposed SS OCF 1 Q-ball support a bound, locally readable clock for 100 calibrated periods?

Experiment: gross.bound-clock.v1 Model: signal-space.ss-ocf-1.flat-clock.v1

Runtime configuration: fixtures/research/gross-test-06.json

Plan lock: 0eeda2dfd7bf4131013bafdfc0fb0e173449a571d79dfa6ca464a20618c4a348

MODEL AND NUMERICAL SETUP

Action: SS OCF 1 flat decoupling action, $U(s)=s-s^2+s^3$, $V_{\chi}(s)=0.25-0.4s+0.2s^2$, $\zeta=0.1$; $\Phi=F(r)\exp(-i\omega_Q t)u_0$, $a=0$.

Units: $c=\hbar=m=1$, (+---); r and t in m^{-1} , fields in m , Ω^2 in m^2 , Q dimensionless.

Domain: Spherical radius $R=80$ with $h=0.1$; trial $\omega_Q/m=0.868, 0.875, 0.90, 0.94$; selected smallest solved bound $E/Q<1$ candidate.

Boundaries: Regular $r=0$, Dirichlet radial $u=r\Phi$ and $w=r\chi$ at R ; outward damping $\gamma=0.08((r-60)/20)^2$ on momenta in outer layer, logged charge/energy sink. $R=160$ independent control with sponge at 140.

Initial Data: Continuous regular BVP with fixed seed radii 5,10,20,40,60 projected by Newton to finite difference grid; $\chi=A f(r)$, $\pi_{\chi}=0$, $\pi_{\Phi}=-i\omega_Q \Phi$, amplitudes $A=0, 0.0005, 0.001, 0.002$.

Deterministic root seed 2026092506 unused by solver. Branch continuation begins from the nearest higher solved trial when fixed seeds fail; no physical parameters or trial frequencies change.

Solver: SciPy solve_bvp and tridiagonal discrete Newton; independent symmetric-tridiagonal radial χ eigenproblem and core L+/L- Hessian screen; second-order kick-drift-kick reciprocal Φ/χ evolution. Nominal 100 periods, other amplitudes 20, $h/2$ and $2R/20$. Documented numerical recovery: if fixed seeds do not converge, continue the last solved core in ω_Q with steps ≤ 0.005 using the same regular BVP, and project to the same stencil; preserve all earlier failures.

Tolerances: Profile residual $<2e-10$; 5 times eigen error margin; mesh/domain and amplitude frequency $<1\%$; local eigenfrequency agreement $<2\%$; nominal mode-energy loss and charge drift $<1\%$; finite sample crossing count ≥ 99 . Resources 650s wall/768MiB/35MiB.

COMPETING PREDICTIONS

usable-clock: A positive nodeless core supplies a mode with $0<\Omega^2<0.25$ that ticks locally and retains $>99\%$ mode energy and core charge for 100 nominal periods.

unusable-clock: No accepted core, no bound mode, unstable core, frequency error or leakage beyond gates prevents a calibrated detector in the tested regime.

CONTROLS

nu-off: test whether trapping coefficient creates the mode; method: set $\nu=0$ in independent finite-box eigenproblem while preserving $\eta=0.2$ and core

unexcited: exclude silent phase as local ticking; method: run $\chi=\pi_{\chi}=0$ for 20 periods; log local χ and stationary core density/projector

amplitude: measure weak-amplitude shifts and backreaction; method: evolve peak $\chi=0.0005, 0.001, 0.002$,

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nominal for 100 periods and others for 20

mesh-domain: separate discretization from box effect; method: solve profile and spectrum at $h/2, R=80$ and $h, R=160$; evolve nominal amplitude for 20 periods on each

PREREGISTERED DECISIONS

profiles (Trial core profiles)

Pass: Four frequencies attempted, selected positive nodeless profile with discrete residual $<2e-10$ and $F(0.75R) < 1e-5$

Fail: Complete finite evidence violates the stated locked inequality or qualitative control.

Unresolved: Required branch, refined solution or saved finite record is absent or nonfinite; do not promote an incomplete branch.

Uncertainty: Mesh error by h versus $h/2$ divided by three plus independent R versus $2R$; deterministic finite window, no statistical confidence.

core-energetics (Charge and core energy)

Pass: Selected $E/Q < 1$; charge over 100 periods assessed in charge-ledger

Fail: Complete finite evidence violates the stated locked inequality or qualitative control.

Unresolved: Required branch, refined solution or saved finite record is absent or nonfinite; do not promote an incomplete branch.

Uncertainty: Mesh error by h versus $h/2$ divided by three plus independent R versus $2R$; deterministic finite window, no statistical confidence.

bound-spectrum (Clock bound eigenvalue)

Pass: $0 < \Omega_{\chi}^2 < 0.25$ and distance from zero and threshold $> 5 \max(|\lambda_h - \lambda_{h/2}|/3, |\lambda_h - \lambda_{2R}|)$

Fail: Complete finite evidence violates the stated locked inequality or qualitative control.

Unresolved: Required branch, refined solution or saved finite record is absent or nonfinite; do not promote an incomplete branch.

Uncertainty: Mesh error by h versus $h/2$ divided by three plus independent R versus $2R$; deterministic finite window, no statistical confidence.

trapping-control (No-well and quiet-core controls)

Pass: $\nu=0$ lowest finite-box eigenvalue ≥ 0.25 ; unexcited χ is identically zero in the measured trace

Fail: Complete finite evidence violates the stated locked inequality or qualitative control.

Unresolved: Required branch, refined solution or saved finite record is absent or nonfinite; do not promote an incomplete branch.

Uncertainty: Mesh error by h versus $h/2$ divided by three plus independent R versus $2R$; deterministic finite window, no statistical confidence.

grid-domain (Independent mesh and domain controls)

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Pass: Relative eigenvalue differences and 20-period local frequency differences from both $h/2$ and $2R < 1\%$
Fail: Complete finite evidence violates the stated locked inequality or qualitative control.
Unresolved: Required branch, refined solution or saved finite record is absent or nonfinite; do not promote an incomplete branch.
Uncertainty: Mesh error by h versus $h/2$ divided by three plus independent R versus $2R$; deterministic finite window, no statistical confidence.

branch-screen (Radial stability screen)

Pass: Adjacent Q slope negative; radial $L+$ has one negative and next positive eigenvalue; radial $L-$ phase and $l=1$ translation eigenvalues within 0.03 of zero; central core peak drifts $< 1\%$ in nominal evolution
Fail: Complete finite evidence violates the stated locked inequality or qualitative control.
Unresolved: Required branch, refined solution or saved finite record is absent or nonfinite; do not promote an incomplete branch.
Uncertainty: Mesh error by h versus $h/2$ divided by three plus independent R versus $2R$; deterministic finite window, no statistical confidence.

local-record (Local zero-crossing frequency)

Pass: ≥ 99.3 signed cycles in a 100 eigenperiod window, ≥ 99 positive-going crossings and relative frequency difference from eigenfrequency $< 2\%$
Fail: Complete finite evidence violates the stated locked inequality or qualitative control.
Unresolved: Required branch, refined solution or saved finite record is absent or nonfinite; do not promote an incomplete branch.
Uncertainty: Mesh error by h versus $h/2$ divided by three plus independent R versus $2R$; deterministic finite window, no statistical confidence.

lifetime (Trapped mode lifetime)

Pass: Absolute projected mode-energy change $< 1\%$ over 100 eigenperiods at peak $\chi=0.001$
Fail: Complete finite evidence violates the stated locked inequality or qualitative control.
Unresolved: Required branch, refined solution or saved finite record is absent or nonfinite; do not promote an incomplete branch.
Uncertainty: Mesh error by h versus $h/2$ divided by three plus independent R versus $2R$; deterministic finite window, no statistical confidence.

amplitude (Amplitude dependence)

Pass: Maximum relative frequency difference among half/base/double amplitude over first 20 periods $< 1\%$
Fail: Complete finite evidence violates the stated locked inequality or qualitative control.
Unresolved: Required branch, refined solution or saved finite record is absent or nonfinite; do not promote an incomplete branch.
Uncertainty: Mesh error by h versus $h/2$ divided by three plus independent R versus $2R$; deterministic finite window, no statistical confidence.

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charge-ledger (Charge after boundary sink)

Pass: Maximum absolute $(Q + \text{outer sink} - Q_{\text{initial}}) / Q_{\text{initial}} < 1\%$ for nominal 100 periods

Fail: Complete finite evidence violates the stated locked inequality or qualitative control.

Unresolved: Required branch, refined solution or saved finite record is absent or nonfinite; do not promote an incomplete branch.

Uncertainty: Mesh error by h versus $h/2$ divided by three plus independent R versus $2R$; deterministic finite window, no statistical confidence.

VISUAL QUESTIONS

core-and-well [evidentiary]: Which trial core supplies a spatial mass well and localized clock eigenfunction?

View: PNG/SVG/PDF with exact CSV and raw NPZ; observables: $F(r)$, $V_{\chi}(F^2)$, $f_{\chi}(r)$

usable-clock: bounded oscillation, independent eigenvalue and converged lifetime

unusable-clock: missing mode, unresolved error or lost local oscillations

Controls: mesh-domain, nu-off; uncertainty: finite radial mesh and box, 20-period refinement/control windows; nonspherical dynamics excluded

Data: plot-data/radial.csv, plot-data/summary.json; transforms: radial fields divided by r ; no smoothing; first 60 units displayed

bound-control [evidentiary]: Does the bound eigenvalue survive resolution controls and disappear without trapping?

View: PNG/SVG/PDF with exact CSV and raw NPZ; observables: E/Q , lowest eigenvalue, continuum threshold

usable-clock: bounded oscillation, independent eigenvalue and converged lifetime

unusable-clock: missing mode, unresolved error or lost local oscillations

Controls: nu-off, mesh-domain; uncertainty: finite radial mesh and box, 20-period refinement/control windows; nonspherical dynamics excluded

Data: plot-data/profiles.csv, plot-data/refinements.csv, plot-data/summary.json; transforms: plot saved eigenvalues directly; no fit

local-trace [evidentiary]: Does a local field trace tick for 100 periods when the unexcited core does not?

View: PNG/SVG/PDF with exact CSV and raw NPZ; observables: χ at $r=h$, zero crossing frequency

usable-clock: bounded oscillation, independent eigenvalue and converged lifetime

unusable-clock: missing mode, unresolved error or lost local oscillations

Controls: unexcited, amplitude; uncertainty: finite radial mesh and box, 20-period refinement/control windows; nonspherical dynamics excluded

Data: plot-data/evolutions.csv, plot-data/summary.json; transforms: display first and last five periods without smoothing; normalize time by independent eigenfrequency

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lifetime-convergence [evidentiary]: Does the mode persist with controlled charge, amplitude, grid and boundary errors?

View: PNG/SVG/PDF with exact CSV and raw NPZ; observables: mode energy fraction, charge balance, frequency convergence

usable-clock: bounded oscillation, independent eigenvalue and converged lifetime

unusable-clock: missing mode, unresolved error or lost local oscillations

Controls: mesh-domain, amplitude; uncertainty: finite radial mesh and box, 20-period refinement/control windows; nonspherical dynamics excluded

Data: plot-data/evolutions.csv, plot-data/summary.json; transforms: project on independently normalized eigenmode; account outer damping sink in charge ledger

RESOURCE LIMITS

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{"max_memory_mb": 768, "max_output_mb": 35, "max_wall_seconds": 650}
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PROHIBITED INFERENCES

Flat spherical lifetime is not full nonspherical stability or a nonlinear Einstein core.

A preselected core and mass well do not derive spacetime or the knot/particle spectrum.

Test 6 does not evaluate neutral reception, recoil or observer-invariant remote records.

A trial profile that does not converge with the bounded BVP is unresolved, not evidence of absence of a Q-ball.